

SECTION B

6 (**short**) The output of a linear system, $y(t)$, is governed by the differential equation

$$\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 4y = f(t)$$

where $f(t)$ is the input. Find and sketch the impulse response of the system. [10]

7 (**long**) The function $f(x)$ is periodic with period 2 and is given by

$$f(x) = 2x - x^2 \quad \text{for } 0 \leq x \leq 2$$

(a) Show that f is an even function of x and explain the implications of this for a Fourier Series representation of f . [5]

(b) Find a Fourier Series representation of f , evaluating the coefficients by direct integration. [10]

(c) Show that, for $-1 < x < 1$,

$$\frac{d^2 f}{dx^2} = 4\delta(x) - 2$$

and find a Fourier Series representation for $\frac{d^2 f}{dx^2}$, evaluating the coefficients by direct integration. [5]

(d) Using the result obtained in part (c), find a Fourier Series representation for f and verify that it agrees with the result obtained in part (b). [10]